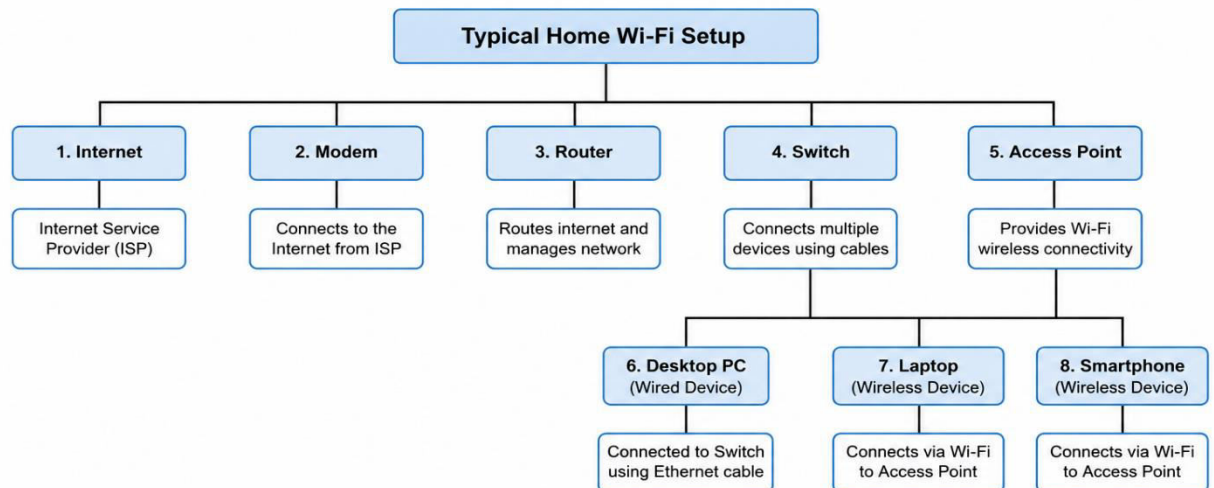


Day 1: Tasks

1. Draw a labeled diagram of a typical home Wi-Fi setup showing a modem, router, switch, access point, and at least two devices (like a laptop and smartphone) connected to the network.



2. List the main function of each networking hardware device: modem, router, switch, access point, and network cable. For each, give a real-life example of where you might see or use it (for example, 'router: at home to share internet').



Networking Hardware Devices and Their Functions

1. Modem

- **Function:** A modem connects a home or office network to the internet by converting signals from the Internet Service Provider (ISP) into a form that computers can use.
- **Real-Life Example:** A modem is commonly installed at home to receive internet service from companies such as JioFiber or Airtel Xstream.

2. Router

- **Function:** A router distributes the internet connection to multiple devices and directs data between them.
- **Real-Life Example:** A router is used at home to share internet access with laptops, smartphones, and smart TVs.

3. Switch

- **Function:** A switch connects multiple devices within the same local network and allows them to communicate efficiently.

- **Real-Life Example:** A switch is often used in offices where many computers need to be connected to the same network.

4. Access Point

- **Function:** An access point provides wireless network access and extends Wi-Fi coverage in an area.
- **Real-Life Example:** Access points are commonly installed in schools, hotels, and large offices to provide Wi-Fi in different rooms.

5. Network Cable

- **Function:** A network cable physically connects network devices and transfers data between them.
- **Real-Life Example:** An Ethernet cable is used to connect a computer directly to a router or switch for a stable internet connection.

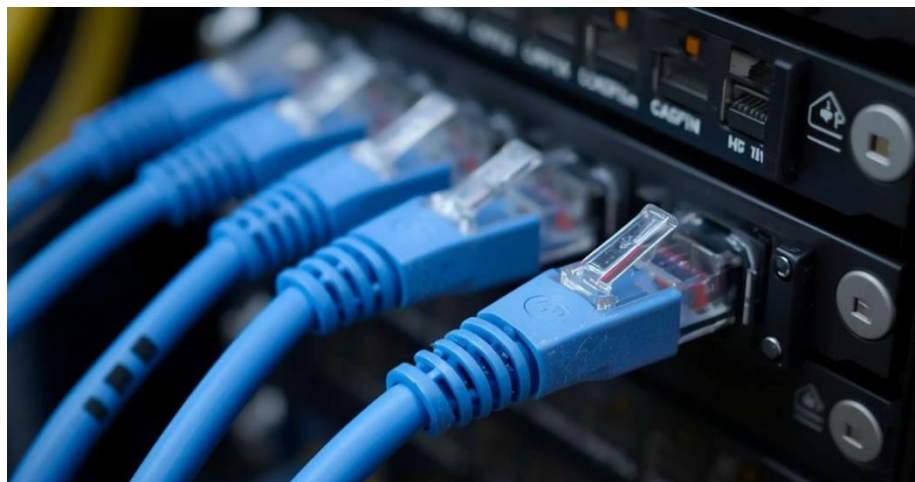
3. Take photos or find images online (with source links) of at least three types of network cables (such as Ethernet, coaxial, fiber optic) and write one line describing where each is commonly used.



Network Cable Types and Their Common Uses

1. Ethernet Cable (Cat5e/Cat6)

- **Common Use:** Used to connect computers, routers, switches, and other devices in home and office networks.
- **Image Source:** <https://www.hkt-enterprise.com/en/blogs/tips-for-business-to-choose-ethernet-cables>



2. Coaxial Cable

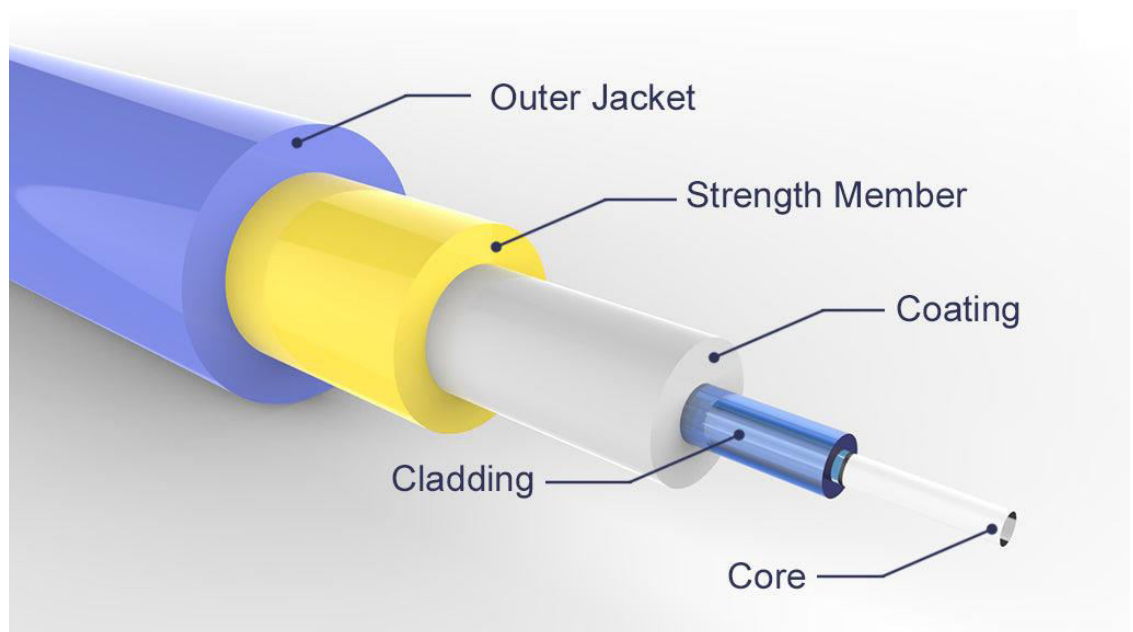
- **Common Use:** Commonly used for cable TV connections, broadband internet, and CCTV camera systems.

- **Image Source:** <https://www.ppc-online.com/blog/what-is-coaxial-cable-and-how-is-it-used>



3. Fiber Optic Cable

- **Common Use:** Used in high-speed internet networks, data centers, and long-distance telecommunications.
- **Image Source:** <https://www.vcelink.com/blogs/focus/fiber-optic-cable-types?srsltid=AfmBOopaFCpSv5qkzvhp0UWzTB9xQkD2RLMItgf5UF4V4z7wjHYABezR>



4. Create a comparison table listing at least four differences between a router and a switch, including their main use, typical location, and how they handle data.



➤ **Comparison Between Router and Switch**

Feature	Router	Switch
Main Use	Connects different networks and provides internet access.	Connects devices within the same network.
Typical Location	Commonly found in homes, offices, and organizations.	Usually found in offices, computer labs, and server rooms.
Data Handling	Sends data between different networks using IP addresses.	Sends data to the correct device using MAC addresses.
Internet Connection	Can connect a local network to the internet.	Cannot directly provide internet access by itself.
Number of Networks	Connects multiple networks together.	Works within a single network.
Example	Home Wi-Fi router.	Office network switch connecting many computers.